

Python Dictionary

1. What is a Dictionary?

A **dictionary** in Python is a collection of **key-value pairs**.

- Keys → unique, immutable
- Values → can be anything (string, int, list, dict, etc.)

Example:

```
person = {  
    "name": "Athar",  
    "age": 30,  
    "role": "Trainer"  
}
```

2. Create a Dictionary

Method 1: Using {}

```
student = {"name": "John", "marks": 85}
```

Method 2: Using dict()

```
student = dict(name="John", marks=85)
```

3. Accessing Dictionary Values

Using the key

```
student["name"] # John
```

Using get()

```
student.get("marks") # 85
```

```
student.get("city", "Not Found") # Default value
```

4. Updating Values

```
student["marks"] = 90
```

```
student["city"] = "Doha"
```

5. Adding New Key–Value Pairs

```
student["course"] = "Data Science"
```

6. Removing Elements

pop() – remove item by key

```
student.pop("marks")
```

popitem() – removes last added pair

```
student.popitem()
```

del – delete key

```
del student["city"]
```

clear – empty dictionary

```
student.clear()
```

7. Looping Through Dictionary

Loop keys

```
for k in student:
```

```
    print(k)
```

Loop values

```
for v in student.values():
```

```
    print(v)
```

Loop both

```
for k, v in student.items():
```

```
    print(k, v)
```

8. Check if Key Exists

```
"name" in student
```

9. Dictionary Methods (Very Important)

Method	Description
dict.keys()	Returns all keys
dict.values()	Returns all values
dict.items()	Returns key-value pairs
dict.update({...})	Updates dictionary
dict.copy()	Returns copy
dict.fromkeys()	Creates dict with default values

Example:

```
keys = ["id", "name", "age"]
```

```
d = dict.fromkeys(keys, None)
```

```
# {'id': None, 'name': None, 'age': None}
```

10. Nested Dictionaries

```
students = {  
    1: {"name": "John", "marks": 90},  
    2: {"name": "Sara", "marks": 88}  
}  
  
print(students[1]["name"]) # John
```

11. Dictionary Comprehension

Pythonic way to build dictionaries:

Example 1: Square numbers

```
squares = {x: x*x for x in range(1, 6)}
```

Example 2: Convert list to dictionary

```
fruits = ["apple", "banana", "grapes"]
```

```
lengths = {fruit: len(fruit) for fruit in fruits}
```

12. Difference Between List & Dictionary

Feature	List	Dictionary
Order	Ordered	Ordered (3.7+)
Access	Index	Key
Speed	Slower lookup	Faster lookup ($O(1)$)
Use case	Sequence	Key-value data

13. Practical Examples

Example 1: Word Count

```
sentence = "python is easy and python is powerful"
```

```
words = sentence.split()
```

```
count = {}
```

```
for w in words:
```

```
    count[w] = count.get(w, 0) + 1
```

```
print(count)
```

Example 2: Student Marks

```
marks = {"John": 80, "Sara": 90, "Ali": 70}
```

```
print(max(marks, key=marks.get)) # Who scored highest
```